

David Bedolla, PhD

Robotics Software Engineer | Real-Time Control | Motion | Hardware Deployment

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PROFILE

Robotics software engineer with applied R&D experience in real-time control, robot dynamics, high-degree-of-freedom systems, and hardware deployment. Experience includes ROS 2 and C++ control systems, redundancy resolution, inverse kinematics, rehabilitation exoskeleton controllers, symbolic modeling, and sensor-based motion generation using IMU, EMG, and depth-camera data. Work connects software architecture, dynamic modeling, motion control, hardware integration, and physical validation for assistive robotics and mobile manipulation.

PROFESSIONAL EXPERIENCE

Associate Researcher

2026-Present | Lab INIT Robots, Montréal

- Deployed 400 Hz whole-body control for an AgileX mobile base and Kinova Gen3 arm, with PS4 commands and validation on the physical system.
- Implemented analytical redundancy resolution for position-level configuration targets in approximately 3 microseconds on an Intel i7 processor.
- Used a Pinocchio-based Jacobian solver and numerical optimization to generate real-time joint-velocity commands for Cartesian tracking and elbow reconfiguration.
- Integrated robot-state feedback, world-frame tracking, joint limits, coordinated base and arm motion, and safety-oriented command logic in ROS 2 and C++.

Postdoctoral Researcher

2025 | ÉTS, Montréal

- Reduced lower-limb robot control latency by 30 times, enabling torque-based control and lower tracking error.
- Improved ROS and Python real-time scheduling for autonomous vehicle control, reducing steering oscillations.
- Developed symbolic dynamics and inverse-kinematics GUI tools for rapid robot modeling and validation.
- Supported international collaborations on wheelchair-mounted robots and upper-limb rehabilitation systems.

Research Professor

2024 | Universidad Tecnológica de la Mixteca, Mexico

- Developed and tested an optimization strategy for a 7-DoF robot, improving tracking performance by 20 percent.
- Supported the mechanical design of a three-finger adaptive robotic gripper for activities of daily living.
- Led a national student race-vehicle project through design, construction, testing, and competition.

Research Assistant

2020-2023 | ÉTS, Montréal

- Deployed closed-form inverse kinematics in approximately 5 microseconds and learning-based inverse kinematics in approximately 43 microseconds.
- Developed a 1 to 4 kHz hard real-time motion-control stack and deployed learning-based predictive control in approximately 387 microseconds.
- Reduced symbolic model-expression size by approximately 90 percent and computed feedforward torque in approximately 210 microseconds.
- Built LabVIEW and Simscape digital twins and integrated EMG, IMU, and depth-camera sensing.
- Designed an EMG and IMU mirror-rehabilitation system for upper-limb therapy.

Research Professor

2019 | Universidad de la Sierra Juárez, Mexico

- Implemented neural-network inverse kinematics for a 6-DoF PUMA robot.
- Developed a MATLAB convolutional neural network for wildlife detection in camera-trap datasets.

Professor

2017-2018 | Tecnológico Nacional de México, Mexico

- Deployed a real-time robotic manipulator emulator on a TI microcontroller with a 10 microsecond sampling period.
- Implemented an FPGA-based voice-processing system using linear predictive coding.

EDUCATION

2020-2023	PhD in Robotics Learning-Based Upper-Limb Robotic Rehabilitation, ÉTS, Montréal
2014-2016	M.Sc. in Robotics Hardware-in-the-Loop Simulation of a Robotic Manipulator, Universidad Tecnológica de la Mixteca
2009-2014	B.Eng. in Mechatronics Universidad Tecnológica de la Mixteca, CENEVAL EGEL Outstanding distinction

SELECTED PUBLICATIONS

- **Bedolla-Martínez, D.**, Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023). *Learning human inverse kinematics solutions for redundant robotic upper-limb rehabilitation*. Engineering Applications of Artificial Intelligence, 126, 106966. DOI: 10.1016/j.engappai.2023.106966.
- **Bedolla-Martínez, D.**, Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023). *Robust MPC with Integral Super-Twisting for Trajectory Tracking of an Exoskeleton Robot Arm*. IEEE PEDS. DOI: 10.1109/PEDS57185.2023.10246735.
- **Bedolla-Martínez, D.** (2023). *Learning-Based Upper Limb Robotic Rehabilitation*. Doctoral dissertation, ÉTS.

SELECTED PROJECTS

- **Whole-Body Mobile Manipulation:** 400 Hz ROS 2 and C++ coordination of a mobile base and Kinova Gen3 arm.
- **Dynamics Forge:** Interactive modeling and control platform for copters, drones, pendulums, cart-pole systems, and manipulators.
- **Automatic Robotics Tooling:** Symbolic kinematics, inverse kinematics, RNEA, and expression-optimization tools for serial and branched robots.
- **Learning-Based Rehabilitation:** Gaussian Process inverse kinematics, robust predictive control, and sensor-driven mirror therapy.

AWARDS AND SCHOLARSHIPS

2024	ÉTS Excellence Award nomination
2020	Doctoral scholarship - National Mexican Council of Humanities, Science and Technology
2014	Master's scholarship - National Mexican Council of Humanities, Science and Technology

TECHNICAL SKILLS

Programming	C, C++, Python, MATLAB, LabVIEW, VHDL
Robotics	ROS 2, Pinocchio, URDF, motion planning, real-time control, mobile manipulation
Modeling and Control	Kinematics, dynamics, numerical optimization, RNEA, MPC, sliding-mode control, Gaussian Processes
Simulation and Hardware	Simulink, Simscape Multibody, LabVIEW RT and FPGA, DAQ, microcontrollers, IMU, EMG, depth cameras